ACS 70
DRIVE- AND
CONTROL SYSTEM

USER MANUAL



6410V0010

Forword with reading instructions

In this user manual You will get a presentation of ACS 70, a drive- and control system especially developed for Albany Door Systems industrial doors. You will get an overall description of how to use, to manage and maintain the system and what you should do in case of failure.

Read the user manual carefully in order to facilitate finding the part You need working with ACS 70.

The user manual has to be placed in the control box in order to be available when needed.

Next is an overview of the manual's content and disposition:

1 Safety regulations

Overall description about the importance to pay attention to the safety regulations and which regulations are valid.

2 General description

Describes ACS 70 and its advantages together with Albany Door Systems.

3 Management

Gives a description on how to use ACS 70 and shows the possibilities one has to change its functions.

4 Maintenance

Contains instructions about constructive maintenance.

5 System outline

Shows the possibilities of ACS 70 standard equipment and available options.

6 Technical data

Appendix

Here You will find the manufacturer's address and space for notes.

Note

The information in the user manual can become out of date because of changes made in newer versions of machine- or programware.

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1. Safety regulations

Current regulations

All the current regulations are according to the enclosed CE-document.

Safety aspects

The Albany Door Systems industrial doors has good safety records, but apart from this, nobody should be in the opening of the door or nearby more than necessary.

Main switch

The power supply should be cut off when working with the door. The main switch should be lockable.

Electrical connections

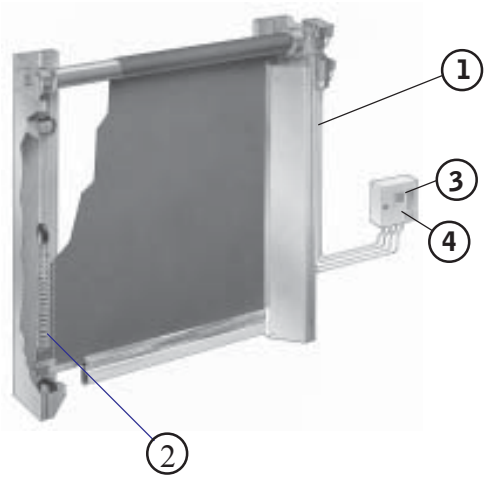
All work with electrical connections or equipments must be done by an electrically skilled person or a person with equivalent knowledge and experience.

Before the work begins, the incoming supply voltage must be cut off with a lockable main switch. After that, the working place is marked with a warning sign that labour is in progress. Then measure the incoming supply with a voltmeter to establish that the control box is out of power.

Note!
That the motor control contains capacitors. Wait with labour at least five minutes after the voltage has been cut off, to let the capacitors discharge.

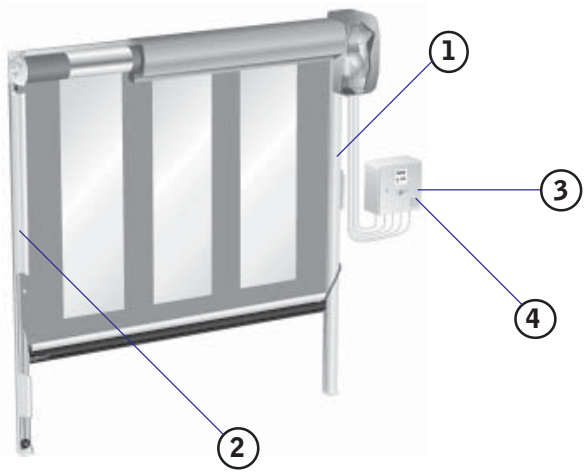
Placement of decals

RapidRoll 400, 600, 900

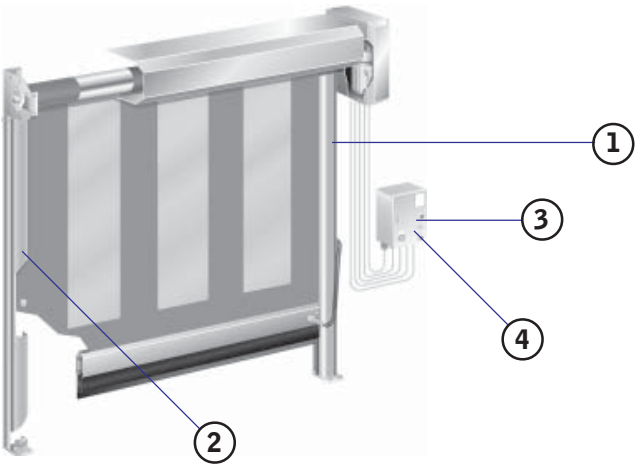


1. Emergency opening decal
2. Model identification
3. Model identification
4. User decal

RapidRoll 330

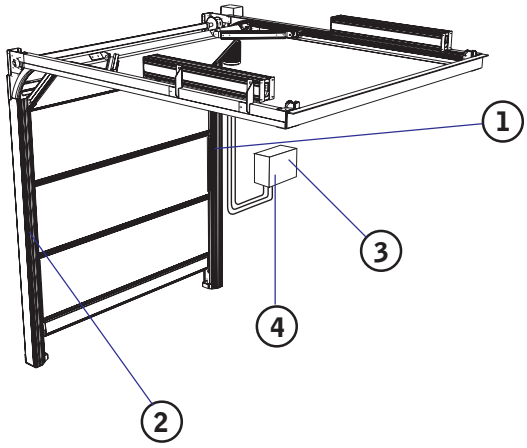


RapidRoll 320



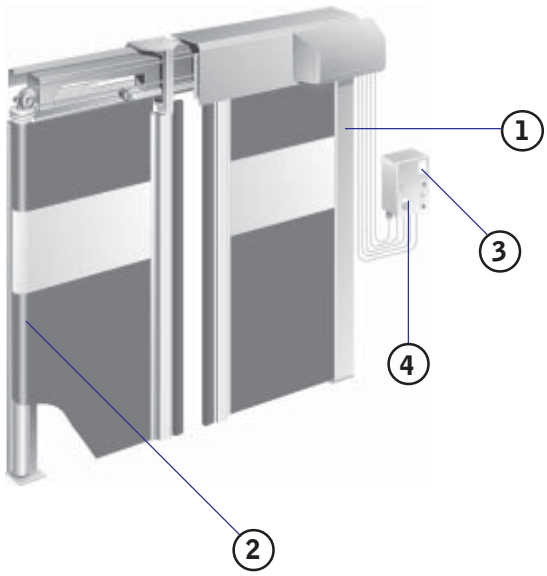
Placement of decals

RapidTech



1. Emergency opening decal
2. Model identification
3. Model identification
4. User decal

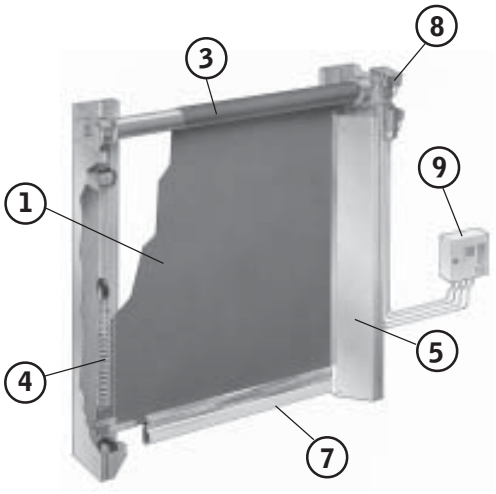
RapidRoll x2



2. General description

The door's components

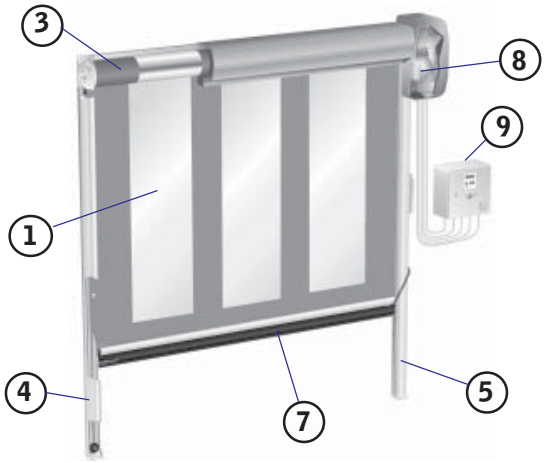
RapidRoll 400, 600, 900



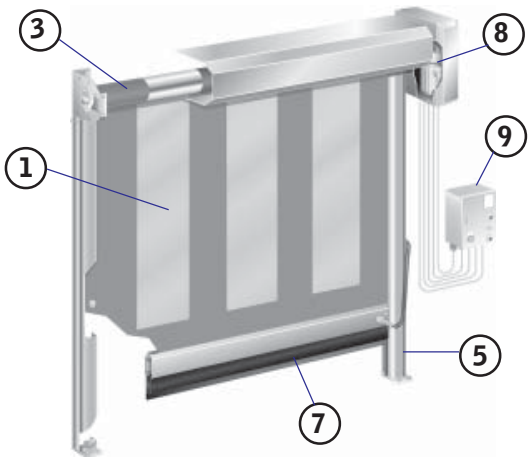
Albany Door Systems RapidRoll consists of a flexible door panel which retracts onto a roll. The door panel is guided at each side by frame sections. These frames incorporate a patented wire and spring system, which stretches the door panel and distributes its weight evenly (valid for door certain door types).

The doors can be used in all types of openings, both indoors and outdoors. Albany Door Systems RapidRoll doors are well-known for their speed. The ACS 70 control and drive system makes it possible to further increase speed – thereby reducing the door's opening time.

RapidRoll 330



RapidRoll 320



RapidRoll comprises the following parts:

(1) Door panel

Coated, colored woven polyester monofilament. Alternative: transparent PVC.

(3) Door roll

Steel/aluminium tube with welded end caps. Shafts and wire drums are bolted onto the caps.

(4) Balancing mechanism

Patented spring system, which balances the door panel throughout its upward and downward movement.

(5) Frame section

Anodized aluminum and/or galvanized steel profiles. Frames completely cover the operating mechanism of the door, for maximum safety and cleanliness, but can be easily opened for inspections, cleaning and maintenance.

(7) Safety edge

On impact, the door reverses immediately to its open position. ACS 70 control- and drive systems are specially developed for Albany Door Systems doors, making optimal control and manoeuvring possible.
Great attention has been given to **personal security**. The safety edge is self-testing, the program is supervised, and motor control is regularly checked by a new patented method. ACS 70 fulfills, by a wide margin, any **safety demands** made by authorities.

(8) Drive system

a) **Electrical motor:** 3-phase induction motor.

b) **Worm gear:** hollow shaft design for direct installation on the roll shaft.

c) **Fixing brake:** to fix the door panel in open or closed position.

d) **Encoder:** to measure the position and direction of the door panel.

e) **Torque arm:** flexible attachment of the drive unit.

(9) Control system

The control system is placed in a control box, and consists of:

a) **Logical unit:** to manoeuvre the door.

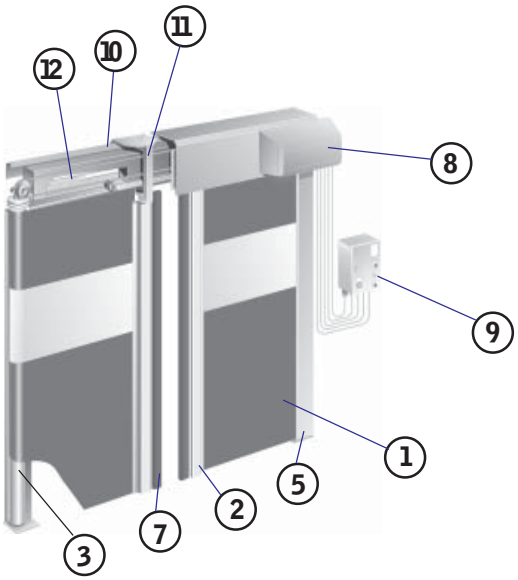
b) **Motor control:** for speed control of the motor.

c) **Panel display:** placed on the front of the control box. Indicates errors by displaying error codes or error messages in plain text. The panel is also fitted with two hold-to-run buttons, to manoeuvre the door in the event of errors such as safety edge errors.

The soft start/stop function means that the door can be run at **extremely high speed** without overloading the transmission, a factor which gives **extended product life**. The need for **service** is **reduced** to an absolute minimum, because slipping and dynamic brake functions are achieved electronically.
The equipment is **noise tested** for external noise, and fulfills European specifications for emission (radio frequency emissions).
The control system is supplied with 1-phase 200, 230 or 240 VAC. If other 1-phase supply voltage is used, the control system must be supplemented with a transformer for current supply voltage. The frequency converter supplies the motor with 3-phase current.

The door's components

RapidRoll x2



The Albany Door Systems RapidRoll x2 incorporates two flexible door blades, each of which rolls onto its respective vertical door roll. Each roll is encased in a frame. A tensioning system ensures even tension on the door blade between the roll and the door blade beam.

The design and construction of the side-roll Albany Door Systems RapidRoll x2 makes it possible to offer extremely high speed throughout the opening and closing cycle. The vision panel, which is always at eye level, gives the operator good control over incoming traffic. This helps to increase flow, improves the work environment and minimizes energy losses.

RapidRoll x2 comprises the following parts:

(1) Door blade

Coated polyester monofilament weave with transparent polyurethane.

(2) Door blade beam

The door blade beam swings out on impact, avoiding damage. It is easily returned to its original position.

(3) Door roll

Each vertical steel door roll rests on roller bearings, and is attached at its upper end to the belt drum.

(5) Frames

Galvanized steel profiles. Frames completely cover the operating mechanisms of the door, for maximum safety and cleanliness, but can be easily opened for inspections, cleaning and maintenance.

(7) Safety edge

On impact, the door reverses immediately to its open position. ACS 70 control- and drive systems are specially developed for Albany Door Systems doors, making optimal control and manoeuvring possible.
Great attention has been given to **personal security**. The safety edge and the program is supervised, and motor control is regularly checked by a new method (patent pending). ACS 70 fulfills, by a wide margin, any **safety demands** made by authorities.

(8) Drive system

a) **Electrical motor:** 3-phase induction motor.

b) **Worm gear:** hollow shaft design for direct installation.

c) **Fixing brake:** to fix the door panel in open or closed position.

d) **Optical encoder:** to measure the position and direction of the door panel.

e) **Torque arm:** for attachment of the drive unit.

(9) Control system

The control system is placed in a control box, and consists of:

a) **Logical unit:** to manoeuvre the door.

b) **Motor control:** for speed control of the motor.

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The equipment is **noise tested** for external noise, and fulfills European specifications for emission (radio frequency emissions).
The control system is supplied with 1-phase 200, 230 or 240 VAC. If other 1-phase supply voltage is used, the control system must be supplemented with a transformer for current supply voltage. The frequency converter supplies the motor with 3-phase current.

(10) Motor beam

Welded construction of steel tube's. Includes attachment points for the tensioning system, frames and drive unit.

(11) Wagon

Caddy wagon, with eccentrically adjustable polyurethane wheels to minimize play and ensure quiet operation.

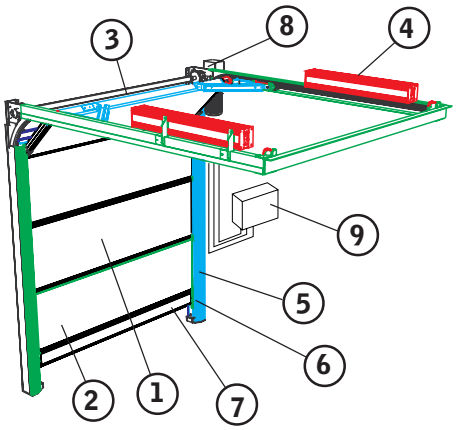
(12) Tensioning system

Tension - springs which ensure that the door blades are fully stretched at each point.

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The door's components

RapidTech



The Albany Door Systems RapidTech comprises a number of rigid panels which retract vertically into a frame in the ceiling when the door is opened.
The Albany Door Systems RapidTech is used for all types of industrial openings where security, durability and speed are a requirement.

RapidTech comprise the following parts:

(1) Panels

Flexible panels of CFC-free extruded polystyrene insulation, with flexible polycarbonate polymer cladding. One or more panels can be fitted with clear polyurethane windows, to provide an exterior view and let in light.

(2) Bottom section

The bottom section is constructed to swing away upon impact, thereby avoiding minor collision damage. The section can be easily snapped back in place.

(3) Transmission shaft

Steel tube with welded end caps. Shafts are bolted into the caps.

(4) Balancing mechanism

Patented spring system, which balances the door panel throughout its movements.

(5) Frames

Anodized aluminum and/or galvanized steel profiles. Frames completely cover the operating mechanisms of the door, for maximum safety and cleanliness, but can be easily opened for inspections, cleaning and maintenance.

(6) Guides

Anodized aluminum rails guide door panel movement. Spring-mounted guide blocks allow panels to run quickly and smoothly, without noise. In the event of a major collision, guide blocks are designed to reduce damage.

(7) Safety edge

On impact, the door reverses immediately to its open position. ACS 70 control- and drive systems are specially developed for Albany Door Systems doors, making optimal control and manoeuvring possible.

Great attention has been given to **personal security**. The safety edge and the program is supervised, and motor control is regularly checked by a new method (patent pending). ACS 70 fulfills, by a wide margin, any **safety demands** made by authorities.

(8) Drive system

a) **Electrical motor:** 3-phase induction motor.

b) **Worm gear:** hollow shaft design for direct installation on the roll shaft.

c) **Fixing brake:** to fix the door panel in open or closed position.

d) **Optical encoder:** to measure the position and direction of the door panel.

e) **Torque arm:** flexible attachment of the drive unit.

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(9) Control system

The control system is placed in a control box, and consists of:

a) **Logical unit:** to manoeuvre the door.

b) **Motor control:** for speed control of the motor.

c) **Panel display:** placed on the front of the control box. Indicates errors by displaying error codes or error messages in plain text. The panel is also fitted with two hold-to-run buttons, to manoeuvre the door in case of errors; for example, safety edge errors.

The soft start/stop function means that the door can be run at **extremely high speed** without overloading the transmission, a factor which gives **extended product life**. The need for **service** is **reduced** to an absolute minimum, because slipping and dynamic brake functions are achieved electronically.
The equipment is **noise tested** for external noise, and fulfills European specifications for emission (radio frequency emissions).
The control system is supplied with 1-phase 200, 230 or 240 VAC. If other 1-phase supply voltage is used, the control system must be supplemented with a transformer for current supply voltage. The frequency converter supplies the motor with 3-phase current.

Safety functions

The doors are equipped with one of the following safety edges:

* Self-testing pneumatic safety edge

* Self-testing optic safety edge

* Continually self-supervised electric safety edge

The optical encoder

The control system receives information on the door's position through the optical encoder. The control system receives also information on the door's moving direction and possible failures.

The inductive encoder

If the drive unit is supplied with an inductive encoder it is also supplied with a limitswitch to indicate a bottom position. For the rest it functions the same way the optical encoder does.

Motion error

Using a patented method, the microprocessor supervises frequency converter, motor and encoder, so that the real motion is what you have selected.

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Emergency stop relay

The emergency stop relay stops the supply to the frequency converter if an error occurs. Each time the door is open the emergency stop relay is tested.

EEPROM

In the event of a power failure, all parameters are saved; e.g. the position in the EEPROM, so that all values will be correct when power supply returns.

Holding brake

Prevents the door panel from sliding down, in the event of damage to the door's balancing mechanism.

Quickstop

In the event of motion errors (see preceding page), the door is stopped immediately.

3. Management

Put on the system

The door is normally ready for use when the installator delivers it.

Parameters as opening- and closing speed, time for automatic closing, opening height etc. are set by the installator.

Sometimes the door is installed at an early stage and is not being used for a certain period in which case the main power supply is off. This is visible because the green diode lamp "**ON**" is off. If this is the case, do as follows:

- Put on the main power break.huvudströmbrytare

- Green diode lamp "**ON**" .

Now the door runs with the help of the push buttons, radio, radar or the liking. The power supply is only to be switched off when under construction. Use the emergency stop if the door is not meant to be running.

Panel display

The panel display is normally placed in the front of the control box and has an display which is switched off in normal operating position. If an error should occur, the **"ERROR"**-indication switches on and an error message is displayed.

At some types of errors the control goes over to hold to run mode. Then the indications above the hold-to-run buttons switch on and you can open and close with these buttons. When the error has been attended to, the indications switch off and you cannot use the buttons any more. If you want to set the control in hold to run mode yourself, press both of the hold-to-run buttons for 2 seconds. The control goes over to normal position when the door has been closed and the safety edge has touched the floor. Hold-to-run mode does not work if emergency stop or crank switch is affected.
If you press the [TEXT] button for 2 seconds the display is switched on and displays the number of door cycles. By pressing the [TEXT] button repeatedly the following is displayed:

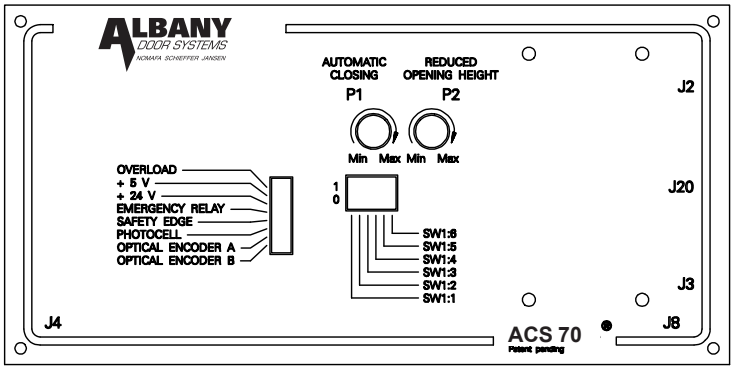
1st time	The position of the door
2nd time	Time for automatic closing
3rd time	Height for reduced opening height in number of pulses
4th time	Opening and closing time
5th time	Frequency
6th time	Set value at upper limit
7th time	Back to the number of door cycles again, etc.

The display switches off again if the [TEXT] button is pressed for 2 seconds (the display should normally be switched off).
The LED "ON" is switched on when the control is in operating position.

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Expanding and tuning of functions

The door is installed with those functions that are specified in the order and order confirmation, but there is an opportunity to change certain parameters according demands.



DIP-switches

SW 1.1 Automatic closing after impact on safety edge.

SW 1.2 The count-down for automatic closing starts when you have passed the photocell in the door-line. If no passage is made, the door closes after 15 seconds.

SW 1.3 Interlocking with other door. Door A must be closed before you can open door B and vice versa.

SW 1.4 Photocell in door-line.

SW 1.5 Flashing traffic light when the door opens and closes, and 2 sec. before automatic closure.

SW 1.6 Only used at installation.

Potentiometers

P 1 Time for automatic closing 1-60 seconds.

P 2 Reduced opening height, Minimum = 400 pulses
Maximum = normal opening height.

Brake release

When needed (f.e. in case of power supply failure), the door can be opened by hand according the following:

Balanced doors (330 with additional balancing, 400, 600, 900):

1. Release the brake by lowering the handle for brake release.

2. Take the bottom beam and raise the door blade to a wanted height.

If the door needs to be closed manually, release the brake as above and lower the door blade to a closed position.

Big balanced doors are manoeuvred manually according the following:

1. Release the brake by lowering the handle for brake release.

2. Place the crank at the back of the motor, crank untill the door blade has necessary height or untill the door is closed.

If the power supply has been off while releasing the brake, the door has to be closed to get even with the floor and the navigation has switched to a normal situation. Has the power supply not been off, it is possible to run the door as usual with the available navigation tools.

Unbalanced doors are manoeuvred manually according following:

1. Place the crank at the back of the motor, crank untill the door blade has necessary height or untill the door is closed.

2. The door can now be restored as to the description mentioned above.

4. Maintenance

RapidRoll and RapidTech

Note!

* All work with electrical connections or equipment must be done by qualified personnel with electrical expertise.
* Before the work begins, the incoming supply voltage must be cut off with a lockable mains switch. After that, the work place must be marked with a warning sign that labour is in progress.Then measure the incoming supply with a voltmeter to establish that the control box is dead.
* Observe that the motor control contains capacitors. Delay labour inside the control box for at least five minutes after the voltage has been cut off, to let the capacitors discharge.
* Don't run the door with the frame sections open.
* Before working on the door, check that the ladders, scaffolds and other equipment are in proper working order, and in accordance with prevailing safety requirements.

Albany Door Systems recommends that maintenance on door take place every 20,000 - 25,000th door opening or every 6 months, according to the maintenance schedule below.

Control and maintenance

RapidRoll, door

* Check bolted joints and, if necessary, tighten the loose bolts.
* Open the frame doors and visually check wires, springs/weights, pulleys and guide rails. If the wires are frayed (watch your hands) they must be changed.

RapidRoll and RapidTech, safety edge

* Check the safety edge with hands, to see if the door returns when touching. This should be done by affecting the safety edge when the door is closing.

RapidRoll and RapidTech, drive unit

* Check the attachment of the drive unit, and tighten loose screws.
* Check shaft packings to ensure that no oil has leaked out. If it has, the gears should be changed. Fill up with new oil, type Mobil SHC 629 or equivalent.
* Check the brake by trying to move the door panel up or down by hand. This should be impossible. After that, disengage the brake with the hand release. It will then be possible to move the door panel up or down.
* Check that the cut-out is functioning, and cut drive current once the crank is in place. This cut-out is positioned at the lower end of the motor near the hole where the fabric is attached.



NOTE!

Spilled or leaking oil can cause accidents. Clean the floor thoroughly.

RapidRoll and RapidTech, other

* Check the control equipment – buttons, photocell, radar, etc. If necessary, clean the lenses of the photocell. Check push buttons, ropes, radar, etc. whether the right function is displayed.

RapidTech, general inspection

A general inspection should be made:

* Guide blocks and guide brackets concerning wear and tear
* Clamp rings and bearings on drive shaft whether they are loosened, if so, tighten them
* Balance chain to the balance spring assembly concerning corrosion, may need lubricant
* Drive chain concerning corrosion, may need lubricant
* Ceiling mounts, check fastening, may need tightening
* When the door is running the drive chain should be stretched all time in opened position, visual control. If this is not the case, contact service.

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RapidRoll x2, rope

* The rope should be changed after 200,000 (±10%) cycles.
  The amount of door cycles can be read in the display, see capital 3, Management.

Note! Contact Albany Door Systems for service.

RapidRoll 330

* Check wear and tear on rope and on the elastic ropes.
* The rope has to be renewed after 200,000 openings (±10%).
* The elastical ropes have to be renewed after 100,000 openings (±10%) or every other year.
* The rope for extra balancing has to be renewed after 200,000 openings (±10%).
* The amount of door cycles can be read in the display, see capital 3, Management.

Note! Contact Albany Door Systems for service.

5. System outline

The Albany Door Systems doors can be provided with a number of different control equipments, such as radar, radio, photocell, inductive loops etc. With ACS 70, the options increase to further adjust the door to your needs. ACS 70 gives you optimal control of the Albany Door Systems door to suit your needs.

Below follows a list on some of the options.

■ Standard.

□ Prepared.

T Option.

Control system

■ Soft start/stopp.

■ Adjustable opening speed.

■ Adjustable closing speed.

■ Adjustable acceleration.

■ Adjustable deceleration.

■ Opening with time-delayed closing.

■ Time-delayed closing after impact on safety edge.

□ Count-down of time for automatic closing when one has passed photocell in the door-line. The door closes after 15 seconds, if nobody has passed.

■ Pause at reversing.

■ Open-stop-close.

■ 2 opening heights.

■ Zero voltage protection.

■ Error indication.

■ Operating voltage 200, 230, 240 V, 50-60 Hz.

T Other operating voltage.

■ Traffic lights with fixed light.

□ Flashing traffic lights.


Manual reset after impact on safety edge.


Hold to run operation.


Interlocking with other door.


Airlock function.


Customer adjusted control program.


Pneumatic safety edge.


Electric safety edge.


Optic safety edge.


Touchless safety edge-STS.


Self tested safety edge.


Photocell in door line.


Extra photocell in door line.


Thermostat in motor.


Manual opening of door in case of power failure (not for all door types).


2 inductive loops.


More than 2 inductive loops.

Control equipment

- Open - close push buttons.
- Emergency stop push button.
- Open-stop-close push buttons.
- Pull switch.
- Detector for opening loops.
- Detector for safety loops.
- Detector for unidirectional loops.
- Photocell in door line.
- Opening photocell.
- Radio.
- Radar.
- Unidirectional radar.
- Red traffic light.
- Red + green traffic lights.

6. Technical data

Technical data

Supply voltage	200, 230 ,240 VAC ±10%
Maximal fuse incoming voltage	10A with 1,1 kW motor 16A with 1,5 kW motor
Internal fuses	F1 - 1A F3 - 1A F4 - 2A
Ambient temperature	-10°C - +50°C
Storage temperature	-40°C - +80°C
Insulation class	Class II with opto switch control circuit and transformer
Noise tested	ESD secure up to 20 kV. EMS secure according VD 0875 (CISPR 14), VDE 0871.

APPENDIX

If You have any questions or need for service, contact your Albany Door Systems representative nearest by.

Manufacturer:

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Box 509
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Kistingevägen 2
S-305 94 Halmstad
Sweden
Tel: +46 - 35 - 147 300
Fax: +46 - 35- 147 399
E-mail: info.ads.se@albint.com
Internet: www.albanydoorsystems.com

NOTES

Date	Number of door openings	Remarks

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Photocell

IMPORTANT. Please read the Safety instructions in User's manual before you start the installation!

General

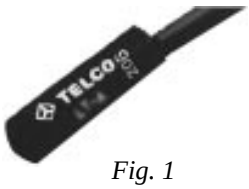


Fig. 1

The photocell have a sensing range of 15 m with Nomadrive and 35m with the PA 10 B 532 amplifier. The amplifier will be desribed in the section Amplifier.

Transmitter

Type	LT 12 A
Connection	Cable 15 m
Size	Ø 10 x 32 mm
Angle	3 °
Sealing	IP 67
Operating temperature	- 25 °C to 65 °C

Receiver

Type	LR A
Connection	Cable 15 m
Size	Ø 10 x 32 mm
Angle	10 °
Sealing	IP 67
Operating temperature	- 25 °C to 65 °C

Amplifier, PA 10 B 532

User Instructions

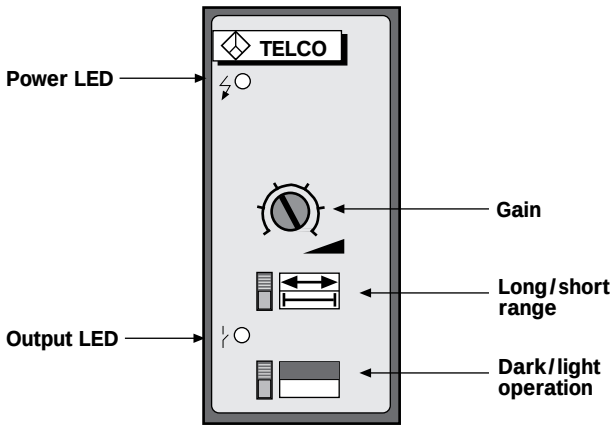


Fig. 2

Instructions

1. Connect wiring to the relay socket as shown in the wiring section.

2. Plug in PA with power OFF.

3. Turn power on after dubble checking your wiring.

4. Select **Light** operation.

5. Select required operation range (long or short - see fig 2).

6. Red LED indicates activated output.

7. Adjust the sensitivity: Turn potentiometer counter clockwise until light diode goes out. Turn clockwise until they lights again. Turn a quater-turn further clockwise. Sensitivity is now optimal. Test the function.

Examples of gain settings

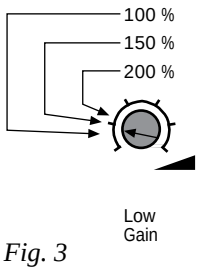


Fig. 3

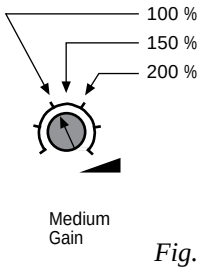


Fig. 4

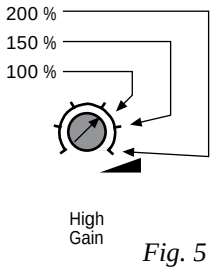


Fig. 5

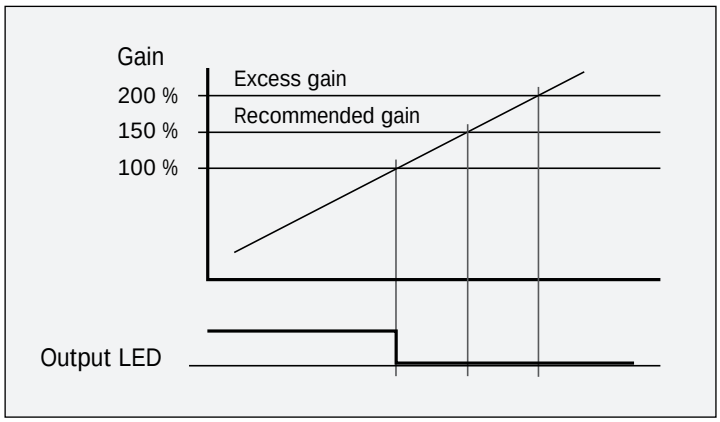
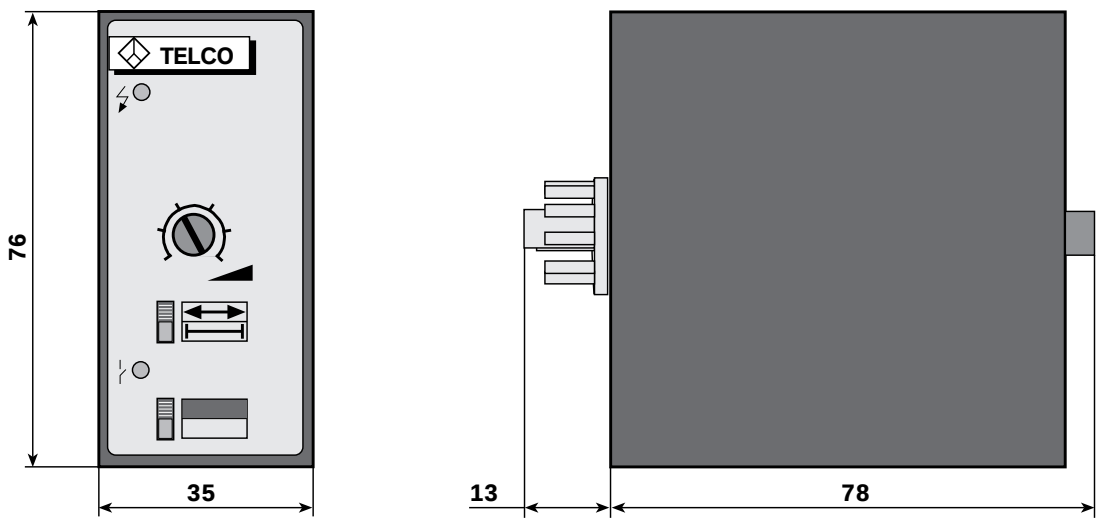


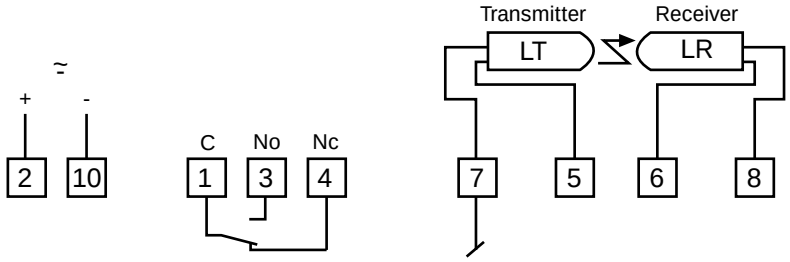
Fig. 6

Dimensions in mm



Technical specifications, Amplifier

Connections



Supply voltage	24 V AC. Other voltages on request.
Voltage tolerance	+/- 15 %
Power consumption	3.2 VA
Output relay	250 V AC/3 A 120 V AC/5 A
Sealing	IP 40

Ordering information

Transmitter/Receiver	Art. No. 96-6410-93
Transmitter/Receiver with 3 m silicon cable	Art. No. 96-6405-94
Amplifier	Art. No. 96-6413-95

Vehicle detector

IMPORTANT. Please read the Safety instructions in User's manual before you start the installation!

General

The loop is made of electrical wire 1.5 mm with a good insulation. Best result is achieved if the loop and the lead-in wire are not joined. If this can not be done it is very important to have a good and waterproof electrical joint.

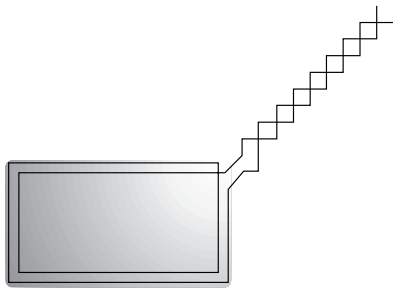
Loops with perimeters up to 12 meters should have 3 turns.

Loops with perimeters exceeding 12 meters should have 2 turns.

Installation

Wire between loop and control box

The lead-in wire is made of the same type of wire as the loop, and it should be twinned at least 10 turns per meter. You can also use a 2-part screen cable. The screen should be grounded in the detector end and not in the loop end.



Size of the loop

The loop size in the traffic direction should be at least twice the distance between the ground and the chassi of the highest vehicle which should traffic the opening, still at least 0.8 m length and 2.0 m width.
To be able to detect bicycles the loop should be 1.1 m long in traffic direction and be laid in a 45° direction against traffic direction.

Where to place the loop

To assure good function, please notice the following;

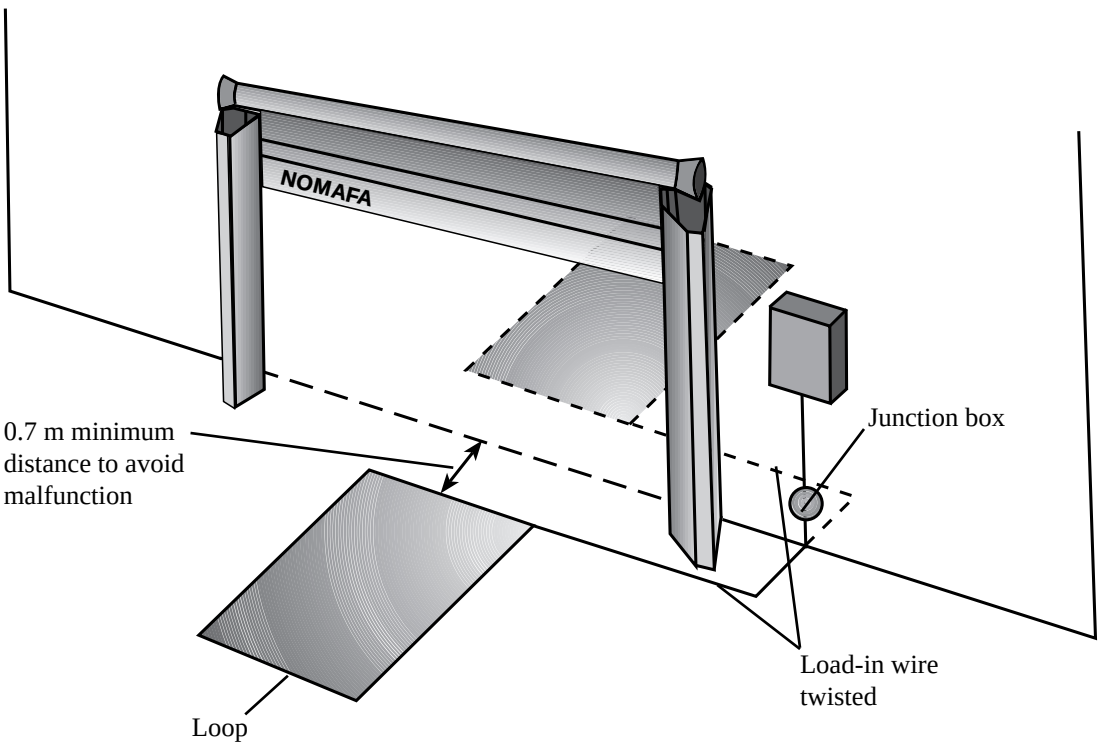
- Loops must be placed an adequate distance from moving doors and other metal objects, at least 0,7 m, to avoid malfunction.

- Minimum distance to reinforcement bars or other solid metal objects: 50 mm.

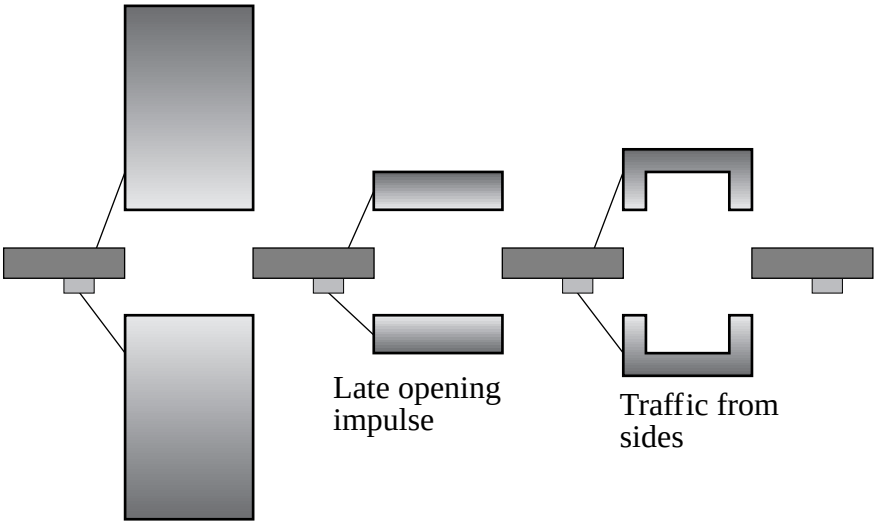
- Connections between the loop, lead-in wire and junctions have to be completely waterproof, also damages on insulation will cause an unstable magnetic field and unsure function.

- Heating cable, high voltage cables, and cables for AGV-trucks, are not allowed to be nearer the loop than 0.5 m.

- The loop track cannot contain other cables.



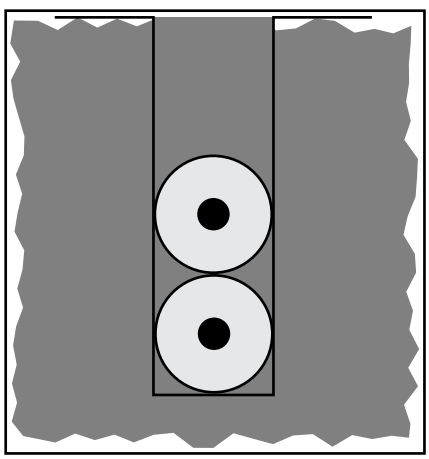
Examples on loop location for different requirements



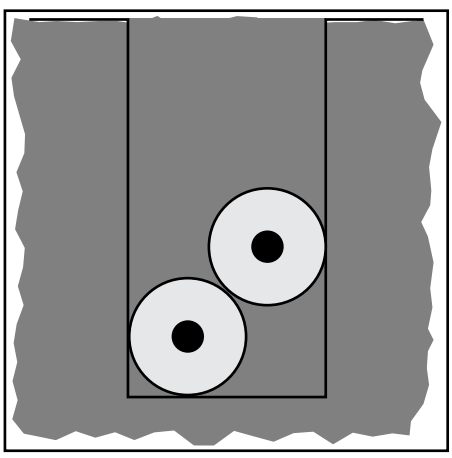
Installing the loop

Loops are normally wound into tracks cut in the road surface. These are typically (5-7 mm Wide) x (35-50 mm Deep) and back-filled with molten asphalt or concrete.

A loop track must not contain any other conductor than the loop itself.
Total feeder length is normally limited to 150 meters.
Loops which are to be installed in reinforced concrete should be as high as possible above the reinforcement bars, at least 50 mm, otherwise loss of sensitivity may occur.



CORRECT placement of loops in track.



WRONG!!

Test the loop

Extreme caution should be exercised when loops are to be installed near any type of heater. The use of loops and heaters together becomes more prohibitive the closer they are laid and impossible if the heating cables passes over or under the loop.

The resistance to earth at 500 V should be greater than 10 MΩ (normally 500 MΩ). Also measure the serial resistance which should be about 0.5 Ω, more than 5 Ω indicate a fault.

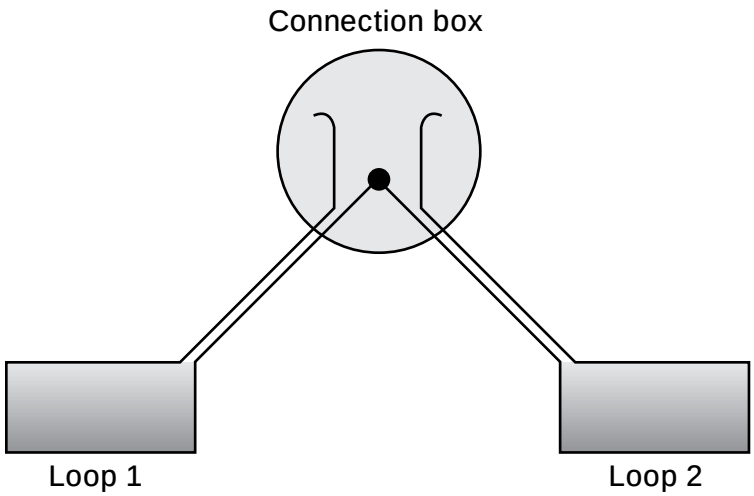
Before backfilling the track, test the system with actual vehicle!

Other

When a 2 loops configuration are operated by one detector, both loops shall be approximately the same size and shape.

The loops shall be connected in series.

When 2 detector systems are operating close to each other be sure to choose different frequency settings.



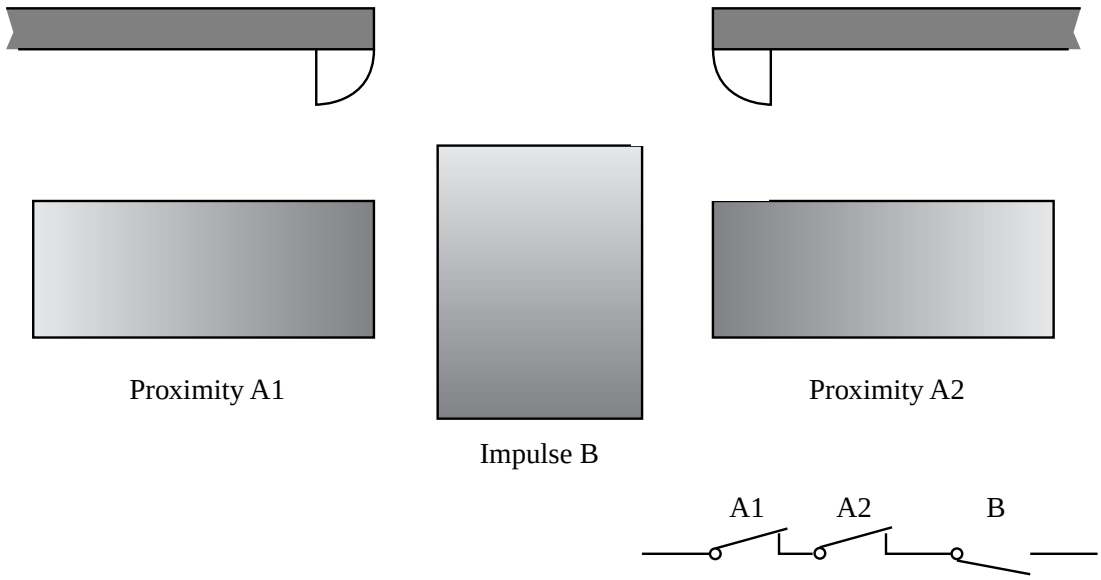
Special coupling loop

Cross-traffic interlock

This arrangement requires two detectors.

Function

The door will not open if loop A1 or A2 actuates before B.



Note!
Only impuls signals from loop B, A1 and A2 should be connected to the same detector.

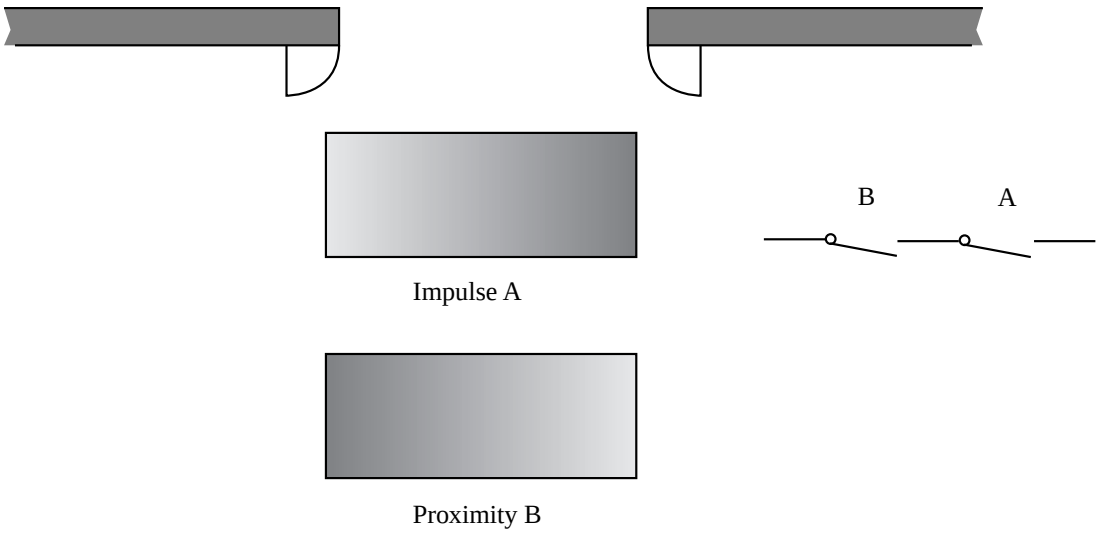
Even though a vehicle is on the loop, the door can begin to close. The special coupling loop should therfore be complemented by a photocell in doorline.

Direction sensitivity loops

This arrangement requires two detectors.

Function

In order to open the door loop B shall be actuated before loop A. Proximity signal loop B, impuls loop A.



Note!
Even though a vehicle is on the loop, the door can begin to close. The special coupling should therfore be complemented by a photocell in doorline.

Description of detector



Features

- Self-contained miniature boxed unit.
- 11 pin relay base connector.
- Presence and pulse output.
- Fast automatic tuning.
- Preset presence time.
- Choice of sensitivity level.
- Comprehensive fault detection.

The use of a microcomputer combines high performance with ease of use in a compact unit. Exceptional noise immunity and reliability enables this unit to function in normally unacceptable conditions.

All detection mode functions are set by means of a front panel rotary switch. Tuning is automatic and fast. Once tuned the detector will track all environmental drifts continuously. The selected presence time is constant and substantially independent of vehicle type. Timing is normally from first vehicle entry.

A loop inductance outside the tuning range of the detector is sensed as a fault. A fault will cause the LED to flash at a 1 Hz rate, with uneven mark-space ratio, and the presence relay to give a permanent presence call. The unit will attempt to retune until the fault is cleared. Whilst retuning the presence call is maintained.

Some vehicles (e.g. high chassis lorries with steel-braced wheels) produce an increase in inductance as they pass over the loop. This will cause some detectors to "lockup". The MZ range of detectors incorporates a feature to prevent this. Normal operation is unaffected.

How to read the loop detector scale

Frequency mode

Used when two detectors are installed close to each other and disturbance between them can appear. If one is switched on high and one on low scale they are not able to react on a wrong signal. If using only one detector it will not make any difference if it is set on high or low.

Presence time

Choose between 5 and 120 minutes. Function is used in case (as example) a forklift charging station is located close to the loop. In this case the door might be activated and stays open until the forklift is moved from the charging place. With this function the detector will accept the forklift as a dead material and close after 5 or 120 min. depending on the selected setting.

Sensitivity

Sensitivity can be made in four steps and depends on the type of forklift. The setting should be tested at place and with the vehicle that should pass.

Examples

Scale no. 3 - High frequence, 5 min, high sensitivity.

Scale no. 4 - High frequence, 120 min, low sensitivity.

Scale no. 12 - Low frequence, 120 min, low sensitivity.

FREQUENCY MODE		**HIGH**		**LOW**	
Presence Time (mins)		**5**	**120**	**5**	**120**
SENSITIVITY	Low	0	*(4)*	8	*(12)*
	Med. low	1	5	9	13
	Med. high	2	6	10	14
	High	*(3)*	7	11	15

Connections

Via 11 pin "socket" at rear of box.

PIN	Function
1	AC/DC +
2	AC/DC -
3	Pulse output, relay common
4	Pulse output, relay contact momentarily closed
5	Presence output, relay contact closed for detect
6	Presence output, relay common
7	Loop input
8	Loop input
9	Ground
10	Presence output, relay contact open for detect
11	Not used

Technical specifications

Tuning range
The tuning range depends on the frequency setting:
- 20 to 1000 µH (high frequency)

Tune time
Normally 2 seconds.

Operating temperature range
- 40° C to 80° C.

Reset
The detector is automatically reset when power is applied or the mode switch is adjusted. If required a reset can be affected by rotating the mode switch one (or more) positions and returning to the required setting.

Response time
Standard response time typically 100 - 150 ms (all sensitivities).

Supply
24 V AC/DC ± 20 % 120 mA max. Other voltages on request.

Relay Outputs
Presence: Changeover relay.

Pulse: Relay contact close for about 100 ms on detection.

Contact: 600 V, 5 A, 250 V AC, 30 V DC.

Dimensions
38 x 76 x 72 mm (W x H x D) excluding connector.

Weight
Approx. 135 g (24 V).

Lightning and transient protection
All inputs are transformer isolated, protected from over voltage on loop leads and flash over from internal circuit to ground.

Ordering information

MZ 24 Art. No. 96-7452-91

MD 1-1
USER'S GUIDE
MOTION SENSORS FOR INDUSTRIAL DOORS
RADAR : for high mounting

TECHNICAL SPECIFICATIONS

Technology : microwave and microprocessor
Transmitter frequency : 24.175 GHz
Transmitter radiated power : < 20 dBm EIRP
Transmitter power density : < 5 mW/cm²
Mounting height : from 3.5 to 7 m
Tilt angle : 0° to 180° in elevation
Detection zone (typical) : 4m (W) x 5m (D) for a mounting height of 5 m
Detection mode : movement
Minimum detection speed : 5 cm/s (measured in the sensor axis)
Supply voltage : 12V to 24V AC +/- 10%
12V to 24V DC +30% / -10%
Mains frequency : 50 à 60 Hz
Power consumption : < 2 W
Output relay : free of potential change-over contact
- Max contact voltage : 42V AC/ DC
- Max contact current : 1A (resistive)
- Max switching power : 30W (DC) / 60 VA (AC)

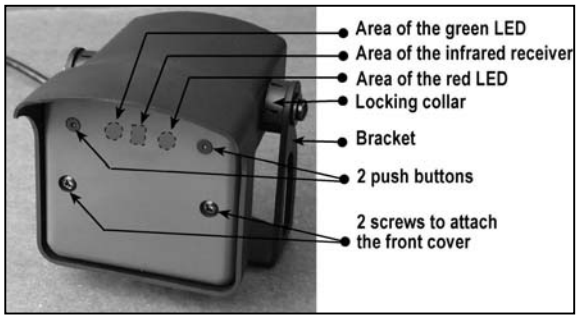
Hold time : 0.5s to 9s (adjustable)

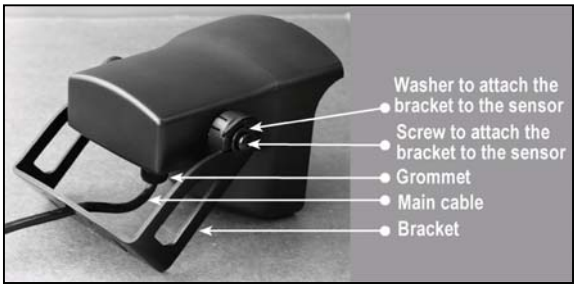
Manual adjustment :
- orientation of sensing field (mechanically)
- multiple functions (by push buttons).

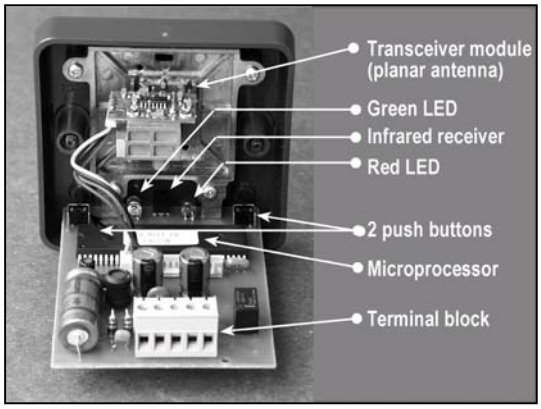
Remote control adjustments :
- Sensitivity.
- Hold time.
- Detection mode.
- Pedestrian and parallel traffic rejection mode.
- Relay configuration.

Temperature range : -30°C à +60°C
Degree of protection : IP65
Product conformity : R&TTE 1999/5/EC
EMC 89/336/EEC
Dimensions : 127mm (D) x 102 mm (W) x 96mm (H)
Weight : 400 g
Plastic of the casing : ABS and Polycarbonate
Bracket : black anodized aluminium
Cable length : 10 m
Cable diameter : 3 mm (minimum)
6.5 mm (maximum)

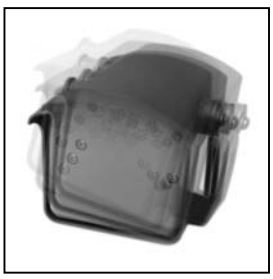
DESCRIPTION OF THE SENSOR







INSTALLATION TIPS



- The sensor must be firmly fastened in order not to vibrate.



- The sensor must not be placed directly behind a panel or any kind of material.

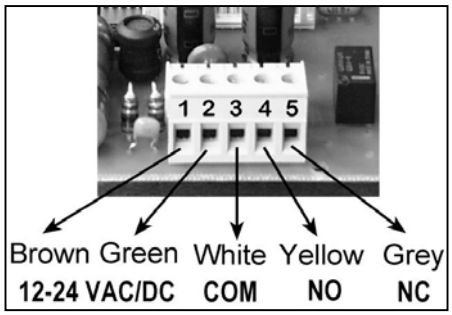


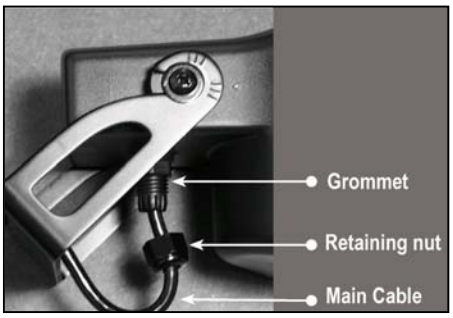
- The sensor must not have any object likely to move or vibrate in its sensing field.



- The sensor must not have any fluorescent lighting in its sensing field.

WIRING



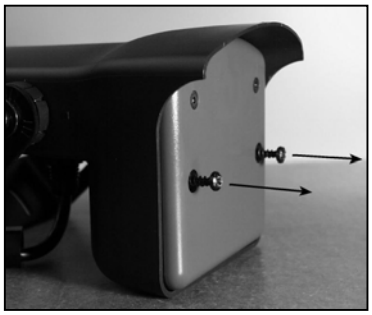


To remove or to insert the cable :

- Unscrew the retaining nut;
- Pass the cable through the grommet and the retaining nut.
- Tighten the retaining nut.

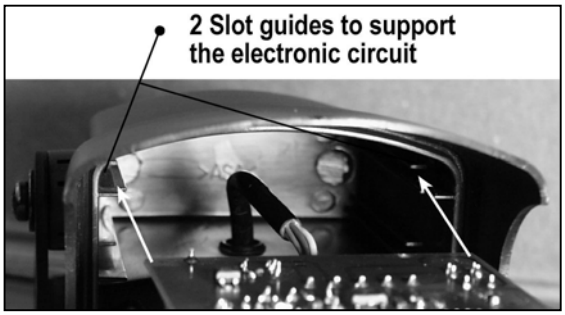
OPENING AND CLOSING THE SENSOR

Opening the sensor



- Loosen the retaining nut until the cable slides easily into the grommet ;
- Partially unscrew the 2 front cover screws ;
- Pull out the front cover with the 2 front cover screws.

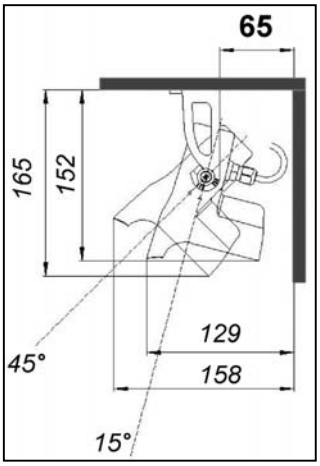
Closing the sensor



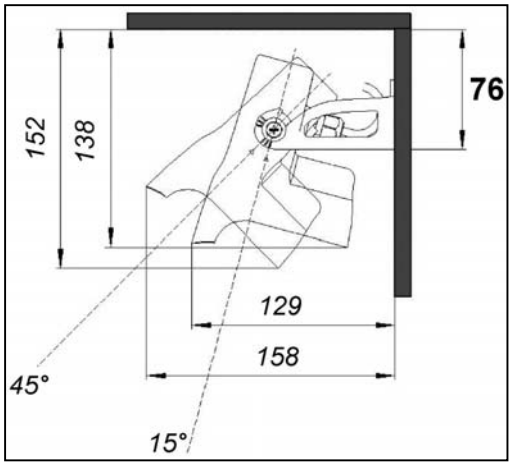
- Connect the quick disconnect terminal block to the main electronic circuit ;
- Slide the main electronic circuit into the 2 housing slot guides and pull the main cable gently at the same time ;
- Gently push the front cover and make sure that the external housing is properly seated (front cover must be flush with housing).
- Screw the 2 front cover screws and tighten the retaining nut.

SENSOR DIMENSIONS AND MOUNTING

Ceiling Mounting

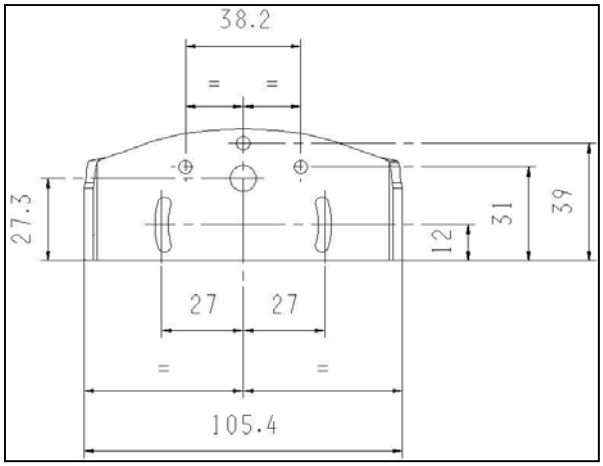


Wall Mounting

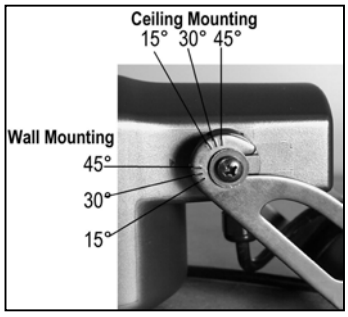


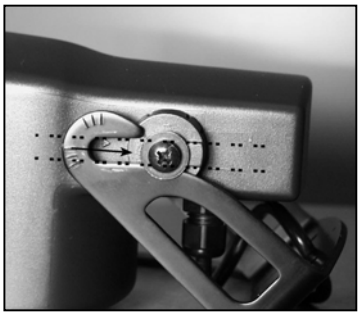
Remark : The bold-type values give the minimum distance required to be able to fully adjust the sensor.

Bracket dimensions



BRACKET MOUNTING



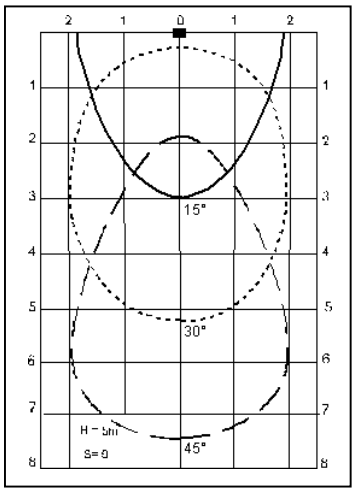


- Check that both locking collar are at the same angle;
- Align the bracket slot to the locking collar guide as shown.

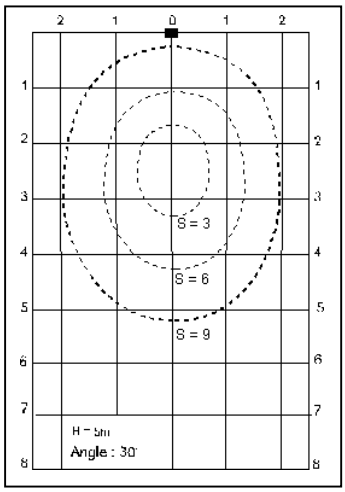
SETTING THE SENSING FIELD DIMENSIONS

MD 1-1 (Mounting height : 5m)

The sensing fields here on the right correspond to the following adjustments :
- tilt angle : 15°, 30°, 45° ;
- sensitivity : 9.

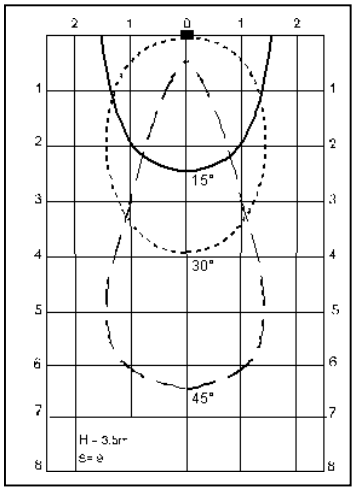


The sensing fields here on the right correspond to the following adjustments :
- tilt angle : 30°
- sensitivity : 9, 6, 3

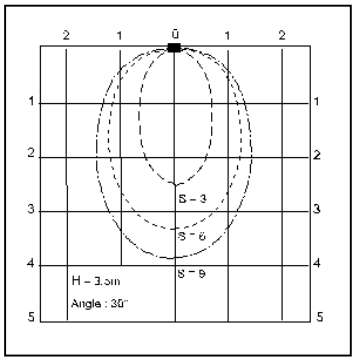


MD 1-1 (Mounting height : 3.5 m)

The sensing fields here on the right correspond to the following adjustments :
- tilt angle : 15°, 30°, 45° ;
- sensitivity : 9.



The sensing fields here on the right correspond to the following adjustments :
- tilt angle : 30°
- sensitivity : 9, 6, 3



LED SIGNAL

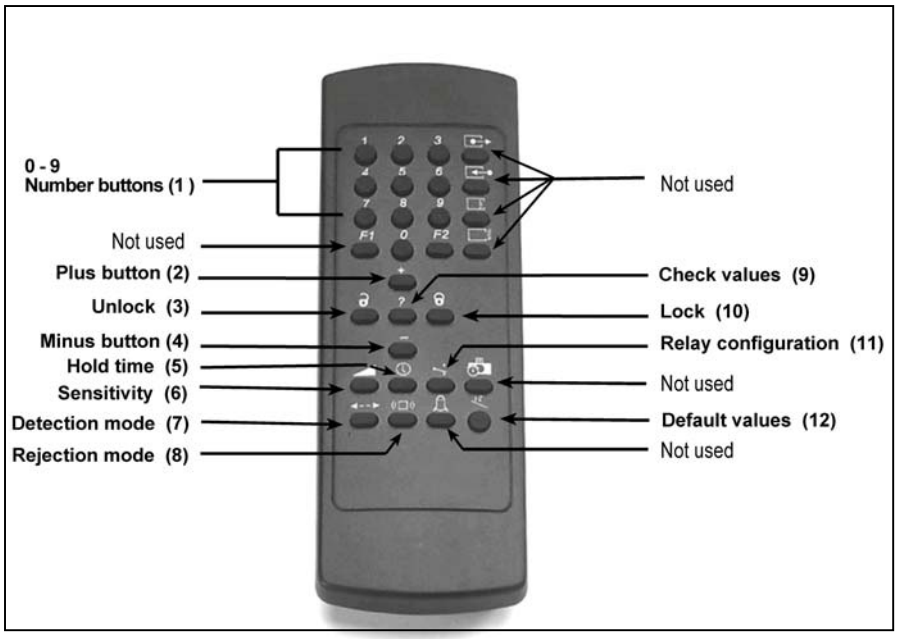
- When the power is turned ON, the red and green LEDs flash for few seconds.
- During a detection the red LED lights on.
- During configuration the red LED flashes a number of times corresponding to the parameter being changed (see next table). The green LED flashes a number corresponding to its setting.

FUNCTIONS CONFIGURATION WITH REMOTE CONTROL

1. DESCRIPTION OF THE INFRARED REMOTE CONTROL



- Open the battery compartment at the back of the remote control ;
- Insert two AAA batteries supplied with the remote control as shown beside ;
- Close the batteries compartment.

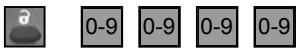


Remark : For optimum results point the remote control at the sensor before pressing its buttons.

2. CONFIGURATION OF THE SENSOR

Each setting change using the infrared remote control must start with an unlocking and end with a locking of the sensor. It is important to point out that any parameters changed using the remote control supersede any previous setting.

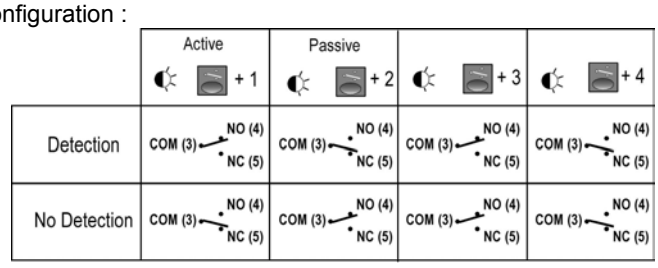
The table below lists all the parameters which can be adjusted with the remote control as well as the operations required to adjust them.

PARAMETER KEY	USER'S ACTIONS	FACTORY SETTING	LED SIGNAL
UNLOCK 	Press the **UNLOCK key (3).** Enter your four-digit access code using **NUMBER keys 0-9 (1)**. During the first sensor adjustment or if the access code is reset to the "0000" value (factory setting) or during the first minute after the power-on, press only the **UNLOCK key (3)** (no code required). *UNLOCK with code*  *UNLOCK without code* 	0000	The red LED flashes quickly waiting for the access code. After entering the correct code or if no code is required, the red LED flashes slowly to indicate that the unlock is successful and the adjustment session has begun. Note : = Adjustment session ON
LOCK 	When all the parameters have been set, press the **LOCK key (10)**. If you wish to enter a new access code, use **NUMBER keys 0-9 (1)** to enter the new four-figure code within 20 seconds. If no access code is entered or if you want to keep the current access code, press the **LOCK key (10)** once more. If no remote control key is pressed within 1 minute, the adjustment session is automatically locked. Pressing the LOCK key (10) twice within the first minute after the power-on reset automatically the access code to 0000 value. *LOCK with code change*  *LOCK without code or Code change* 	0000	The red LED stops flashing to return to its normal function.

Note : All parameters or functions listed in the following tables are only accessible if the sensor is in adjustment session. The RED LED is then slowly flashing.

During an adjustment session each parameter may be checked or changed at any time in the following way :	
PARAMETER KEY	**USER'S ACTIONS**
CHECK VALUES	Press the **key (5,6,7,8,11)** corresponding to the parameter to be checked and then press the **CHECK VALUES key (9)**. Count the number of times the green LED flashes, which corresponds to the value of the checked parameter. No green LED flash corresponds to the value 0. Repeat this operation to check the value of the other parameters if required. Example : SENSITIVITY key (6) – 7 flashes of the green LED : the sensitivity is set at the value 7. *CHECK VALUES* : X
PLUS	Press the **key (5 or 6)** corresponding to the hold time or sensitivity parameter to be modified. and then press the **PLUS key (2)** to increase the value by 1 unit. *PLUS* :
MINUS	Press the **key (5 or 6)** corresponding to the hold time or sensitivity parameter to be modified. and then press the **MINUS key (4)** to reduce the value by 1 unit. *MINUS* :

During a sensor adjustment session all the parameters may be reset to their factory values in the following way :	
PARAMETER KEY	**USER'S ACTIONS**
DEFAULT VALUES	Press the **DEFAULT VALUES key (12),** then press the **NUMBER key 1**. All the parameters are reset to the factory values (see below). *DEFAULTS VALUES :* 1

PARAMETER KEY	**USER'S ACTIONS**	**FACTORY SETTING**	**LED SIGNAL**
SENSITIVITY	Press the **SENSITIVITY key (6).** Use the **NUMBER keys 0-9 (1)** to enter the sensitivity required (or adjust this sensitivity using the **PLUS (2)** or **MINUS (4)** keys as explained above) *SENSITIVITY :* 0-9	7	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.
HOLD TIME	Press the **HOLD TIME key (5).** Use the **NUMBER keys 0-9 (1)** to enter the required hold time (0.5 s to 9 s) (or adjust this parameter using the **PLUS (2)** or **MINUS (4)** keys as explained above). *HOLD TIME* : 0-9	0.5 s	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.
RELAY CONFIGURATION	Press the **RELAY CONFIGURATION key (11).** Use the **NUMBER keys 1-4 (1)** to select the required relay configuration : 	1 (Active Output)	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.
DETECTION MODE	Press the **DETECTION MODE key (7).** Use the **NUMBER keys 1-3 (1)** to select the required mode : key 1 : bidirectional key 2 : unidirectional approach key 3 : unidirectional depart *DETECTION MODE :*	2 (Unidirectional Forward)	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.

REJECTION MODE 	Press the **REJECTION MODE key (8).** **Perturbations immunity** is used to avoid detection due to environmental perturbations (vibrations, rains, etc). **"Pedestrian/parallel traffic rejection"** provides both rejection of pedestrian and rejection of any parallel traffic **simultaneously**. Use the **NUMBER keys 1- 5 (1)** to enter the required rejection mode : key 1 : detection of all kind of targets in motion key 2 : detection of all kind of targets in motion + perturbations immunity key 3 : Low "Pedestrian/parallel traffic" rejection + perturbations immunity key 4 : Mid "Pedestrian/parallel traffic" rejection + perturbations immunity key 5 : High "Pedestrian/parallel traffic" rejection + perturbations immunity The discrimination between a pedestrian and the different vehicles depends mainly on the mounting height and the microwave module tilt angle. Be careful that the rejection function increases the response time of the sensor. Use the next table as suggestions and do not hesitate to increase or decrease the rejection level to obtain the required rejection.	1 (No rejection)	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.

Recommended key		Tilt angle		
		15°	30°	45°
Mounting height	7 m	3	4	4
	5 m	3	4	4
	3.5 m	4	4	5

FUNCTIONS CONFIGURATION WITH PUSH BUTTONS

Without remote control all of the parameters can be set using the 2 push buttons.
Note : These two buttons are accessible from the sensor front cover with a small point.

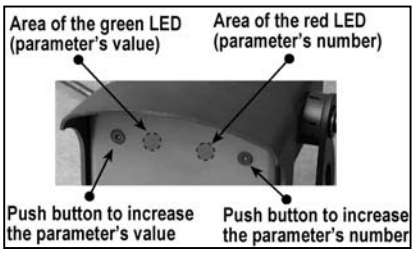
- To adjust the sensor using the buttons :
  Press and hold either push button for 2 seconds (until the LEDs flash) and then release the button ;
- To end adjusting the sensor using the buttons :
  Press and hold either push button for 2 seconds (until the LEDs stop flashing) and then release the button;
  Note : if no button is pressed within 20 seconds the adjustment session is automatically ended.
- To reset all the parameters to the factory values :
  Press and hold both push buttons simultaneously until the two LEDs switch on for 1 second (after 2 seconds).

During the manual adjustment session the red and green LEDs flash successively and continuously.
- The flashing number of the **red LED** provides the **number of the displayed parameter** (see next table);
- The flashing number of the **green LED** provides the **value of the displayed parameter** ;

Note : no red flashing indicates the zero value.

During the manual adjustment session :
- Each press on the **right button increases the number of the displayed parameter** by one unit ;
- Each press on the **left button increases the value of the displayed parameter** by one unit.

Note : When the maximum value or the highest number of the parameter is reached, it will return to its minimum values.



Parameter Number	Parameter	Values	Factory setting
1	Sensitivity	0 - 9	7
2	Hold time	0 - 9	0
3	Relay configuration	1 - 4	1
4	Detection mode	1 - 3	2
5	Pedestrian rejection mode	1 - 5	1

For example, to change the sensitivity from 7 to 9 and the rejection mode from “detection of all kind of targets in motion” to High “Pedestrian/parallel traffic” rejection.

- Press any button for 2 seconds to enter the adjustment session and then release it.
- The red LED flashes once (parameter 1 = sensitivity) and the green LED flashes 7 times (sensitivity=7).
- Press the left button twice to increase the sensitivity from 7 to 9.
- The red LED still flashes once (parameter 1 = sensitivity) but the green LED flashes 9 times now (sensitivity=9).
- Now press the right button 4 times to move to function 5 (rejection mode) ;
- The red LED flashes 5 times (parameter 5 = rejection mode) and the green LED flashes once (“detection of all kind of targets in motion"
- Press the left button 4 times to set the parameter to High “Pedestrian/parallel traffic” rejection.
- The red LED still flashes 5 times (parameter 5 = rejection mode) but the green LED flashes 5 times now (High “Pedestrian/parallel traffic” rejection).
- Press any button during 2 seconds to end the adjustment session and then release it.

TROUBLE-SHOOTING

SYMPTOM	***PROBABLE CAUSE***	***CORRECTIVE ACTION***
The door will not open and the red LED does not light up.	The sensor power is off.	Check power supply. Check the supply voltage.
The door opens and closes constantly.	The sensor "sees" the door moving.	Increase the tilt angle and/or reduce the sensitivity.
	When closing the door creates vibrations picked up by the sensor.	Ensure that the sensor is correctly fixed. If the rejection mode is set at level 1, set this parameter to level 2. Reduce the sensitivity. Switch to unidirectional mode.
The door opens and closes after a given time for no apparent reason.	The sensor is picking up unintended traffic motion.	Reduce the sensitivity. Reduce the tilt angle.
The sensor is not capable of activation near the door.	The tilt angle is too large.	Reduce the tilt angle.
The sensor does not respond to the remote control.	The batteries are weak.	Check the batteries insertion. Change the batteries.
	The access code has been changed.	• Press both push buttons simultaneously to reset all the parameters to the factory values. • Or switch off the power supply. Within the first minute after the power on, change the access code.

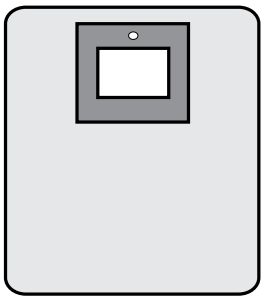
Radio

IMPORTANT. Please read the Safety instructions in User's manual before you start the installation!

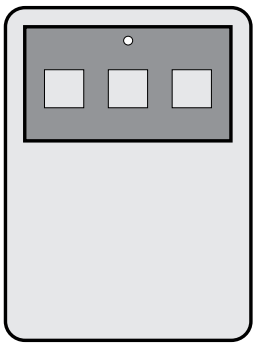
General

The system transmitts at a frequency of 40 MHz in the AM band. Telecoms reserves these frequencies for this type of short-distance transmissions. A radio signal is sent from the hand- or the vehicle- transmitter to a receiver located close to the rapid roll door. A relay in the reciever activates and the door opens.

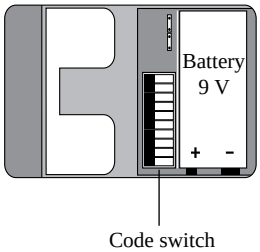
Hand transmitter



One push button for opening of one door.
Type 201

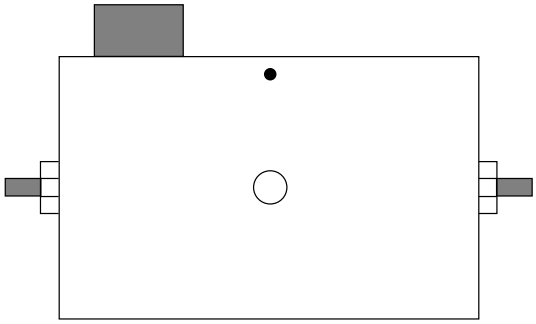


Three push buttons for opening of three different doors or one door with UP - STOP - DOWN.
Type 203

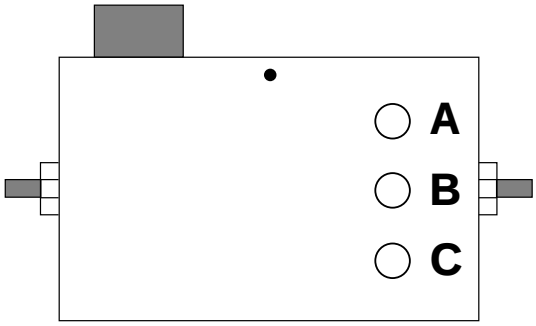


Transmitter is supplied from a 9 V battery placed under a cover on the rear side of the hand transmitter. Life time of battery at normal use is between 3 - 12 month.

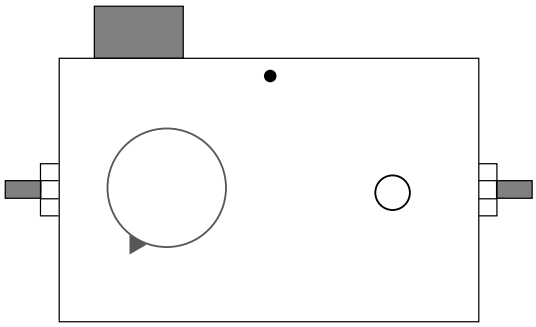
Vehicle transmitter



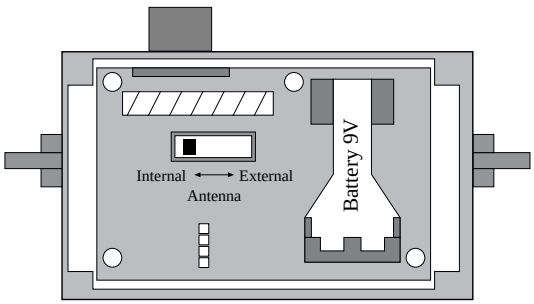
One push button for opening of one door with impulse function.
Type S300 1B



Three push buttons for opening of three different doors or one door with
UP - STOP - DOWN.
Type S300 3B

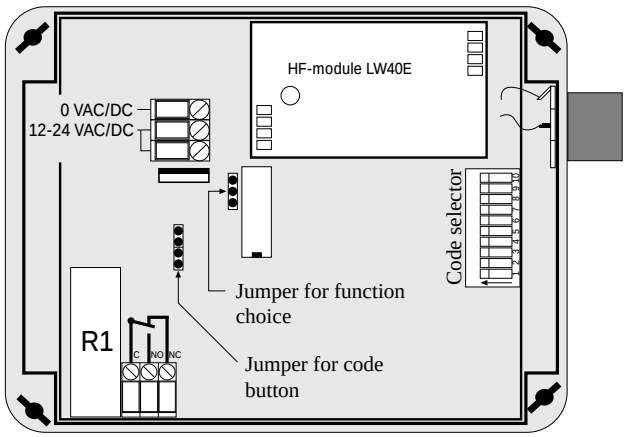


One push button and a rotary switch from 1 - 10 permitting control of up to 10 doors.
Type S300 3B

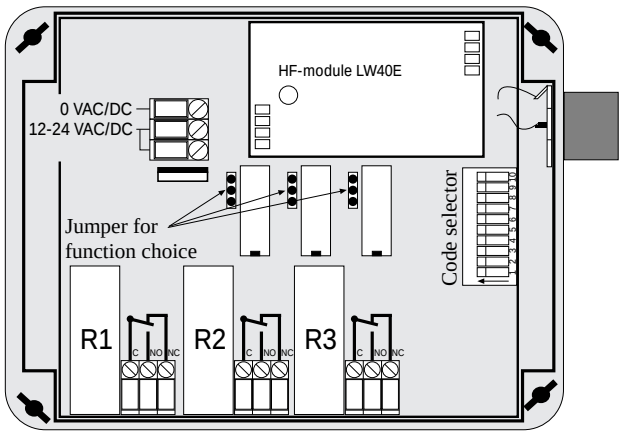


Transmitter is supplied from a 9 V battery placed under the front cover. Life time of battery at normal use is between 3 - 12 month. This transmitter is also available for external 10 - 35 V DC power supply.

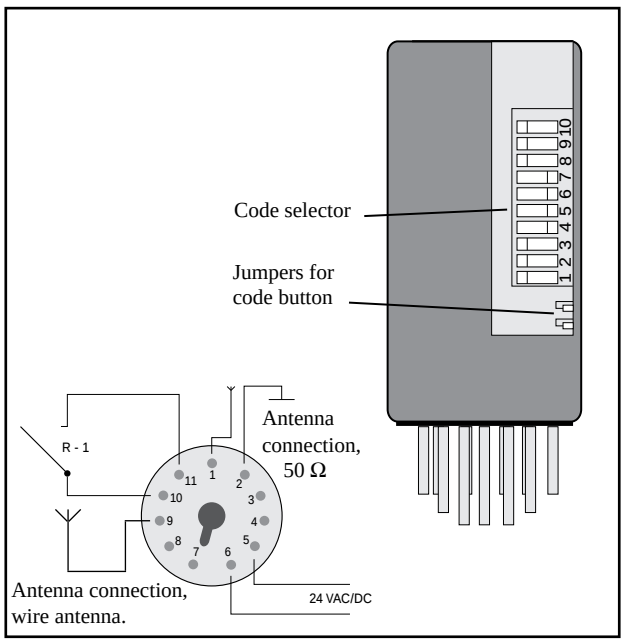
Receivers



One channel external receiver
Type 7344
(24 V AC/DC)
One relay output.



Three channel external receiver
Type 7346
(24 V AC/DC)
Three relay output.



One channel internal receiver
Type 7314, (24 V AC/DC). All connections are made on a standard 11-pole socket. One relay output.

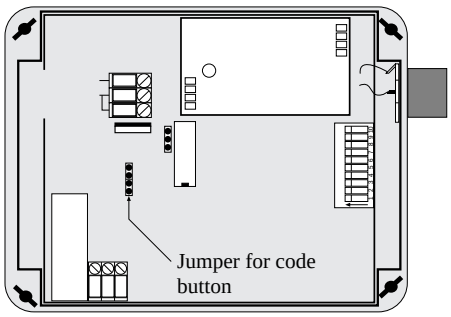
The antenna can be connected in two different ways:
1. Coaxial cable connected to No. 1 and the screen to No. 2.
2. Wire antenna or coaxial cable connected to No. 9 (the screen should not be connected).
Note! With the antenna connected according to item 2, it´s possible to have longer operating range but the receiver will be more sensitive for disturbances.

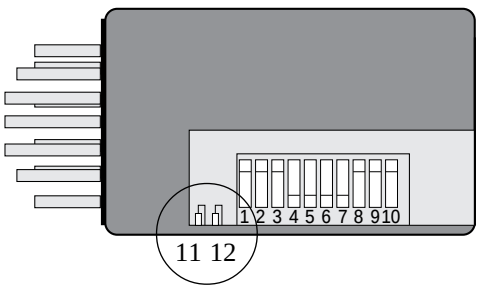
Coding the transmitter and receiver

The radio system must be coded at the time of installation, so that the system's transmitters and receivers all function logically. Consider these examples. The first applies for multi-channel receivers, the second for single-channel receivers.

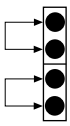
Multi-channel receivers
Three receivers, numbered ONE, TWO and THREE, mounted in three separate locations. With a 10-channel transmitter, the respective receivers should be coded so that receiver ONE functions when the transmitter's dial is set in one-position, and so on.

Single-channel receivers
Four single-channel receivers, and a three-button transmitter. In this case, transmitter and receiver code switches are set identically. Only the respective receivers' jumpes are adjusted. As a result, each receiver will function for a different button on the transmitter. Illustrated below are the settings for the receiver's codes for respective transmitter buttons.

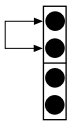




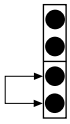
Jumper position for selection of transmitter button to the 1st receiver.



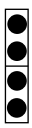
Jumper position for selection of transmitter button to the 2nd receiver.



Jumper position for selection of transmitter button to the 3rd receiver.



Jumper position for selection of transmitter button to the 4th receiver.



Connect both jumpers 11 and 12 to receive signals from the 1st button on the transmitter.

Connect jumper 11 to receive signals from the 2nd button on the transmitter.

Connect jumper 12 to receive signals from the 3rd button on the transmitter.

Disconnect both jumpers 11 and 12 to receive signals from the 4th button on the transmitter.

Code key

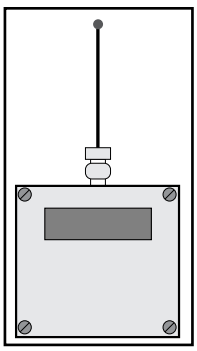
10-channel rotary switch vehicle transmitter and hand transmitter

Decide which receiver should function when the transmitter is set to position '1', position '2', and so on.
Each transmitter contains a 6-pole code switch which determines the user's security code. Set one or more of the switches to the ON position at the transmitter's code selector. These settings must be identical on all the system's transmitters.
Remove the cover from each receiver, exposing the 10-pole code selector on the printed circuit card. Set switches 1-6 in the same positions as on the transmitters. (Example: transmitter switches 1 and 5 are set to ON. Therefore, set receiver switches 1 and 5 to ON, and 2, 3, 4 and 6 to OFF.)
Security codes are now set. Now we shall set receiver codes to function with the transmitter's different channels. Using this code key, up to ten different 3-channel receivers can be set to function with one or more multi-channel transmitters.
Use a two-digit combination to number the different receivers from 01 to 10, corresponding to different transmitter channels. Then set respective receivers' code switches to match the transmitters' channel selector positions, as illustrated below.

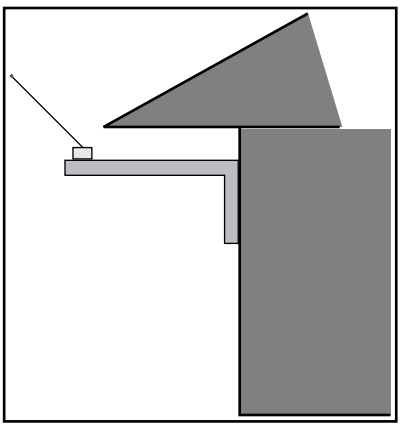
Rotary switch position: (transmitter channel)	**Switch set to ON position**	**Switch set to OFF position**
1	10	7, 8, 9
2	9	7, 8, 10
3	9, 10	7, 8
4	8	7, 9, 10
5	8, 10	7, 9
6	8, 9	7, 10
7	8, 9, 10	7
8	7	8, 9, 10
9	7, 10	8, 9
10	none	7, 8, 9, 10

Antennas and their position

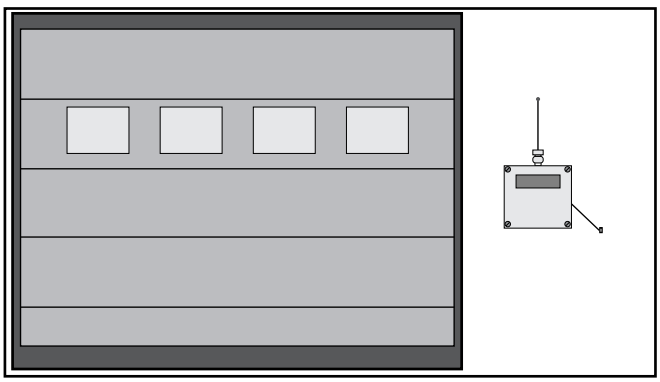
In a radio control system, all receivers should be equipped with some kind of antenna. It is the receiver's antenna which, principally, decides the radio installation's range.
The vehicle's transmitter should also have an antenna if it's placed in an area with a lot of obstructions. Inside the transmitter there is a switch for internal or external antenna. As a rule, one could say that the transmitter and the receiver should "see" each other. If they can't, one of them has to be relocated. The antenna should have an angle of 45 °. It is important that the antenna has a good earth connection.
Here are a few different types of antenna positions:

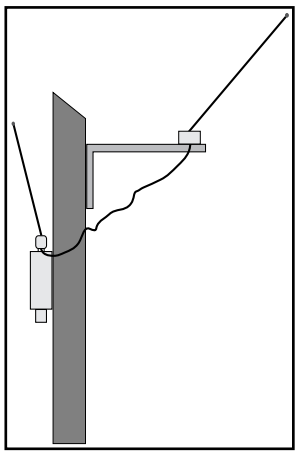


Antenna mounted directly on the wall receiver or directly on the control box if the receiver is built in.



Antenna with cable mounted on a wall with a bracket.





Wall receiver equipped with an inner and an outer antenna.

Technical specifications

Transmitter	
Frequency	40.685 MHz
Amplified modulated pulses	
Power supply hand type	9 V
Power supply mobile type	9 V or external 10 - 35 V DC
Power consumption	45 mA
Receiver	
Sensitivity	1.5 µV at 10 dB
Power supply	24 V AC/DC. Other voltages on request.
Power consumption	25 mA
Relay output	250 VAC, 10 A

Ordering information

Hand transmitters	**Type 201,** Art. No. 96-7360-85 **Type 203,** Art. No. 96-7362-85
Vehicle transmitters	**Type S300 1B,** Art. No. 96-7362-85 **Type S300 3B,** Art. No. 96-7368-85 **Type S300 10B,** Art. No. 96-7369-85
Receivers	**Type 7344,** Art. No. 96-7373-85 **Type 7346,** Art. No. 96-7377-85 **Type 7314,** Art. No. 96-7371-89